

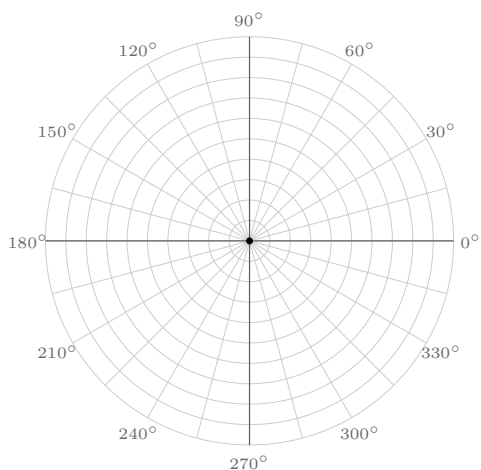
Polar Unit Test (Practice A)

*Calculator***SHOW ALL WORK FOR FULL CREDIT.****You have the entire period.**

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. Show your set up (what you are putting into the calculator). (3 points each)

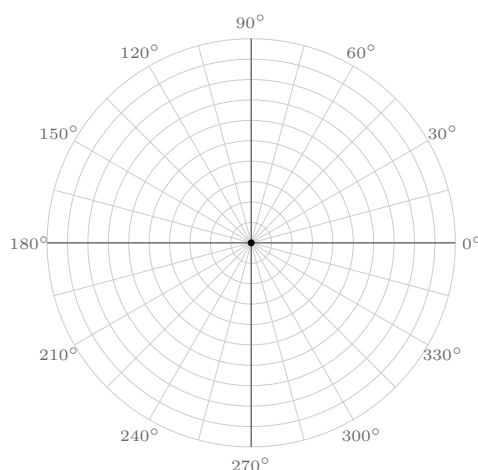
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 3 + 6 \cos \theta$



Type: _____

2. $r^2 = 25 \sin 2\theta$



Type: _____

Polar Unit Test (Practice A)

*Non-Calculator***SHOW ALL WORK FOR FULL CREDIT.****Simplify. (#3 is 3 points and #4 is 2 points)**

3. $\frac{\sqrt{10}}{\sqrt{-2} \cdot \sqrt{-6}}$	4. $\frac{-6 + \sqrt{-72}}{9}$
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Perform the indicated operations. Write answers in rectangular form. (2 points each for 5 & 6 and 3 points each for 7 & 8)

5. i^{99}	6. $(-4 - 3i) - (-6 + 2i) - (9 + 5i)$
7. $\frac{\sqrt{10} + i\sqrt{10}}{\sqrt{5} - i\sqrt{5}}$	8. $2i(3 - i)(1 + 4i)$

Solve over the set of complex numbers. (3 points)

9. $2x^2 + 5 = -2x$

Convert to trig form. (3 points)

10. $2 - 2\sqrt{3}i$

Convert to rectangular form. (3 points)

11. $6 \operatorname{cis} \frac{5\pi}{6}$

Find the product. Write it in rectangular form, using exact values. (3 points)

12. $[\sqrt{3} \operatorname{cis} 145^\circ] [\sqrt{3} \operatorname{cis} 185^\circ]$

Using $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$ is the following in the Julia set? Show ALL work. (3 points)

13. $z = 1 + i$

YES / NO

Convert to trig form, then divide. Write the quotient in rectangular form. (3 points)

14. $\frac{-4\sqrt{3} - 4i}{1 + i\sqrt{3}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(1 + i\sqrt{3})^5$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

16. $x^3 + 27 = 0$

The fractal called the Mandelbrot set: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does not belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

a) $z = -1$

YES / NO

b) $z = 1$

YES / NO

For the point given in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is not an equivalent point, and (c) find the rectangular coordinates of the point in exact form. (3 points)

18. $(-6, 300^\circ)$

a) $(6, 120^\circ)$

b) $(-6, -60^\circ)$

c) $(6, -240^\circ)$

d) $(6, 300^\circ)$

Give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$. (3 points)

19. $(-2\sqrt{3}, -2)$

Convert each rectangular equation to a polar equation. (2 points each)

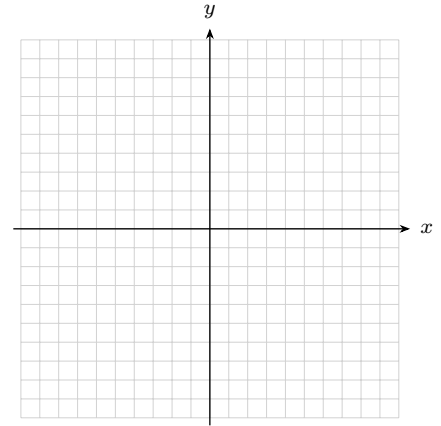
20.

a) $3x - 2y = 6$

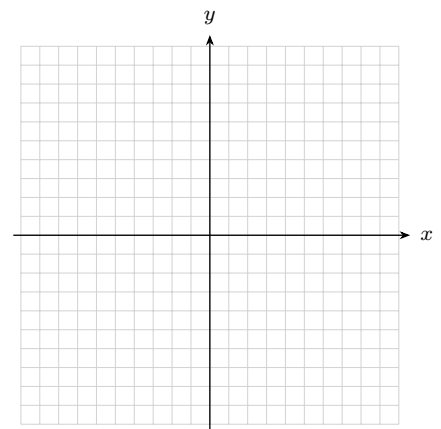
b) $x^2 + y^2 = 49$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (3 points each)

21. $r = 6 \cos \theta$



22. $r = \frac{4}{1 + \sin \theta}$



Find the polar equation of the line. Simplify your answer. (3 points)

23. Passes through $(6, 0^\circ)$ and $(4, 90^\circ)$

Find the polar coordinates of the points of intersection for the given interval of θ . (3 points)

24. $r = 2 + 2 \cos \theta$, $r = 3$
 $0 \leq \theta < 2\pi$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

a) $r = 5 \sin 3\theta$

b) $r = 4 + 4 \cos \theta$

c) $r^2 = 36 \cos 2\theta$

d) $r = 2 + 5 \sin \theta$

e) $r = 7$

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

Signature

ANSWER KEY – Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

1. $r = 3 + 6 \cos \theta$

WORK

Polar mode: $r1(\theta) = 3 + 6 \cos \theta$

$\theta = 0^\circ: r = 3 + 6(1) = 9$

$\theta = 90^\circ, 270^\circ: r = 3$

$\theta = 180^\circ: r = 3 - 6 = -3$

$\frac{a}{b} = \frac{3}{6} = \frac{1}{2} < 1$

limaçon (inner loop)

WHY

Degree or radian mode both work; a blank screen means the mode or window is off.

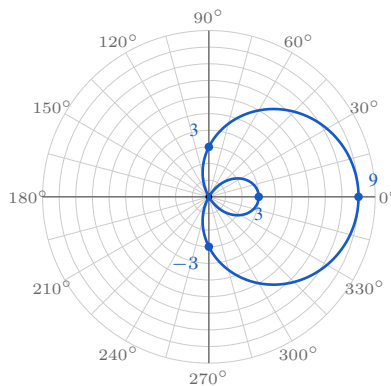
Graph Trace reads r . Rightmost point $(9, 0)$.

Crosses the y -axis at 3 and -3 .

Negative r walks backward: inner loop ends at $(3, 0)$.

Ratio under 1: loop inside.

Form $a + b \cos \theta$ with $a \neq b$.



Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

2. $r^2 = 25 \sin 2\theta$

WORK

$r = \pm \sqrt{25 \sin 2\theta}$

$r1(\theta) = \sqrt{25 \sin(2\theta)}$

$\theta = 45^\circ: r^2 = 25(1), r = 5$

$\sin 2\theta < 0$ in QII and QIV

lemniscate

WHY

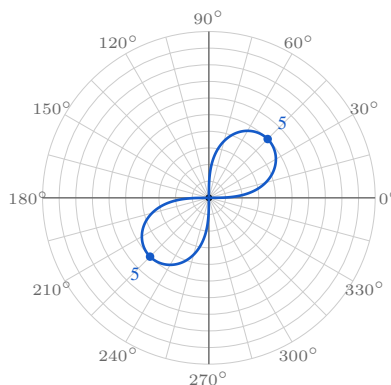
The calculator needs $r =$; either root gives the same graph.

Enter this one; the $-$ root retraces it.

Tip of the loop in QI; 225° gives the QIII tip.

No graph there: loops only in QI and QIII.

r^2 equation, infinity shape.



ANSWER KEY – Non-Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Simplify. (3 points)

3. $\frac{\sqrt{10}}{\sqrt{-2} \cdot \sqrt{-6}}$

WORK

$$\begin{aligned}\frac{\sqrt{10}}{i\sqrt{2} \cdot i\sqrt{6}} &= \frac{\sqrt{10}}{i^2\sqrt{12}} \\ &= \frac{\sqrt{10}}{-2\sqrt{3}} \\ &= \frac{\sqrt{10}}{-2\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{30}}{-6} \\ &= \boxed{-\frac{\sqrt{30}}{6}}\end{aligned}$$

WHY

Pull out i first, then multiply.

$$i^2 = -1, \sqrt{12} = 2\sqrt{3}.$$

Rationalize: multiply top and bottom by $\sqrt{3}$.

Trap: Rationalize by multiplying by the radical over itself, not by a whole number.

Simplify. (2 points)

4. $\frac{-6 + \sqrt{-72}}{9}$

WORK

$$\begin{aligned}\frac{-6 + 6i\sqrt{2}}{9} \\ &= -\frac{6}{9} + \frac{6\sqrt{2}}{9}i = \boxed{-\frac{2}{3} + \frac{2\sqrt{2}}{3}i}\end{aligned}$$

WHY

$$\sqrt{-72} = i\sqrt{36 \cdot 2} = 6i\sqrt{2}.$$

Split, then reduce each fraction by 3.

Trap: Reduce both parts by the same factor; the i part keeps its own coefficient.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

5. i^{99}

WORK

$$\begin{aligned}99 \div 4 &= 24 \text{ R } 3 \\ i^{99} &= i^3 = \boxed{0 - i}\end{aligned}$$

WHY

Only the remainder matters.

Perform the indicated operations. Write the answer in rectangular form. (2 points)

6. $(-4 - 3i) - (-6 + 2i) - (9 + 5i)$

WORK

$$\begin{aligned}-4 - 3i + 6 - 2i - 9 - 5i \\ \text{real: } -4 + 6 - 9 = -7; \text{ imaginary: } -3 - 2 - 5 = -10 \\ \boxed{-7 - 10i}\end{aligned}$$

WHY

Distribute each minus sign to both parts.

Trap: $-(9 + 5i)$ is $-9 - 5i$: both signs flip.

Perform the indicated operation. Write the answer in rectangular form. (3 points)

7. $\frac{\sqrt{10} + i\sqrt{10}}{\sqrt{5} - i\sqrt{5}}$

WORK

$$\frac{\sqrt{10} + i\sqrt{10}}{\sqrt{5} - i\sqrt{5}} \cdot \frac{\sqrt{5} + i\sqrt{5}}{\sqrt{5} + i\sqrt{5}}$$

top: $\sqrt{50} + i\sqrt{50} + i\sqrt{50} + i^2\sqrt{50} = 2i\sqrt{50} = 10i\sqrt{2}$
 bottom: $5 + 5 = 10$
 $\frac{10i\sqrt{2}}{10} = \boxed{0 + i\sqrt{2}}$

WHY

Multiply by the conjugate of the bottom.

$\sqrt{50} = 5\sqrt{2}$; the reals cancel.
 $a^2 + b^2$.

Perform the indicated operations. Write the answer in rectangular form. (3 points)

8. $2i(3 - i)(1 + 4i)$

WORK

$$(6i - 2i^2)(1 + 4i) = (2 + 6i)(1 + 4i)$$

$$= 2 + 8i + 6i + 24i^2$$

$$= 2 + 14i - 24 = \boxed{-22 + 14i}$$

WHY

Distribute the $2i$ first; $-2i^2 = +2$.

FOIL.

Solve the equation over the set of complex numbers. (3 points)

9. $2x^2 + 5 = -2x$

WORK

$$2x^2 + 2x + 5 = 0$$

$$x = \frac{-2 \pm \sqrt{4 - 4(2)(5)}}{2(2)} = \frac{-2 \pm \sqrt{-36}}{4}$$

$$= \frac{-2 \pm 6i}{4}$$

$$\boxed{x = -\frac{1}{2} \pm \frac{3}{2}i}$$

WHY

Zero on one side first.

Quadratic formula, $a = 2$, $b = 2$, $c = 5$.

$\sqrt{-36} = 6i$.

Split into standard form.

Convert to trig form. (3 points)

10. $2 - 2\sqrt{3}i$

WORK

$$r = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$$\theta = \tan^{-1}\left(\frac{-2\sqrt{3}}{2}\right) = \tan^{-1}(-\sqrt{3}) = -60^\circ$$

$$\theta = -60^\circ + 360^\circ = 300^\circ$$

$$\boxed{4 \text{ cis } 300^\circ}$$

WHY

$r = \sqrt{x^2 + y^2}$.

$\tan^{-1}$ only gives QI or QIV.

$x > 0$, $y < 0$ is QIV: add 360° .

$r \text{ cis } \theta$.

Convert to rectangular form. (3 points)

11. $6 \text{ cis } \frac{5\pi}{6}$

WORK

$$6\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$$

$$= \boxed{-3\sqrt{3} + 3i}$$

WHY

$\frac{5\pi}{6}$ is QII: cos negative, sin positive.

Distribute the 6.

Find the product. Write the product in rectangular form, using exact values. (3 points)

12. $[\sqrt{3}(\cos 145^\circ + i \sin 145^\circ)] [\sqrt{3}(\cos 185^\circ + i \sin 185^\circ)]$

WORK

$$\begin{aligned} & \sqrt{3} \cdot \sqrt{3} \operatorname{cis}(145^\circ + 185^\circ) \\ &= 3 \operatorname{cis} 330^\circ \\ &= 3 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) \\ &= \boxed{\frac{3\sqrt{3}}{2} - \frac{3}{2}i} \end{aligned}$$

WHY

Product theorem: multiply the r 's, add the θ 's.
Only the sum has to be a unit-circle angle.
 330° is QIV: sin negative.
Distribute.

Using $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... is the following in the Julia set? Show ALL work. (3 points)

13. $z = 1 + i$

YES / NO

WORK

$$\begin{aligned} (1 + i)^2 - 1 &= 1 + 2i + i^2 - 1 = -1 + 2i \\ |-1 + 2i| &= \sqrt{(-1)^2 + 2^2} = \sqrt{5} \approx 2.24 > 2 \\ \boxed{\text{NO}}, z = 1 + i &\text{ is not in the Julia set} \end{aligned}$$

WHY

First step.
Exceeds 2.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

14. $\frac{-4\sqrt{3} - 4i}{1 + i\sqrt{3}}$

WORK

$$\begin{aligned} r_1 &= \sqrt{48 + 16} = 8 \\ \theta_1 &= \tan^{-1}\left(\frac{-4}{-4\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ \\ \theta_1 &= 30^\circ + 180^\circ = 210^\circ \\ r_2 &= \sqrt{1 + 3} = 2, \theta_2 = \tan^{-1}(\sqrt{3}) = 60^\circ \\ \frac{8 \operatorname{cis} 210^\circ}{2 \operatorname{cis} 60^\circ} &= 4 \operatorname{cis} 150^\circ \\ &= 4 \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = \boxed{-2\sqrt{3} + 2i} \end{aligned}$$

WHY

Top first.
Calculator-style answer is QI.
 $x < 0, y < 0$ is QIII: add 180° .
Bottom is QI, no fix.
Quotient theorem: divide the r 's, subtract the θ 's.
 150° is QII.

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(1 + i\sqrt{3})^5$

WORK

$$\begin{aligned} r &= \sqrt{1 + 3} = 2, \theta = \tan^{-1}(\sqrt{3}) = 60^\circ \\ (2 \operatorname{cis} 60^\circ)^5 &= 2^5 \operatorname{cis}(5 \cdot 60^\circ) \\ &= 32 \operatorname{cis} 300^\circ \\ &= 32 \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \boxed{16 - 16\sqrt{3}i} \end{aligned}$$

WHY

QI, no fix.
De Moivre: power on r , multiply θ .
 300° is QIV.

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

16. $x^3 + 27 = 0$

WORK

$$\begin{aligned} x^3 &= -27 = 27 \operatorname{cis} 180^\circ \\ \sqrt[3]{27} &= 3 \\ \alpha &= \frac{180^\circ + 360^\circ k}{3} = 60^\circ + 120^\circ k \\ &3 \operatorname{cis} 60^\circ, 3 \operatorname{cis} 180^\circ, 3 \operatorname{cis} 300^\circ \\ &\boxed{\frac{3}{2} + \frac{3\sqrt{3}}{2}i, -3, \frac{3}{2} - \frac{3\sqrt{3}}{2}i} \end{aligned}$$

WHY

-27 sits on the negative real axis.
nth root theorem: root of r .
 $k = 0, 1, 2$: three roots, 120° apart.
Unit circle values times 3.

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

WORK

a) $-1, (-1)^2 + (-1) = 0, 0^2 + (-1) = -1$

b) $1, 1^2 + 1 = 2, 2^2 + 1 = 5$

Trap: 2 does not exceed 2. Keep going one more step.

WHY

Back to -1 : it cycles $-1, 0, -1, \dots$

No absolute value ever exceeds 2.

$|5| = 5 > 2.$

For the point given in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is not an equivalent point, and (c) find the rectangular coordinates of the point in exact form. (3 points)

18. $(-6, 300^\circ)$

WORK

(a) face 300° (QIV), walk backward:

(b) $(-6, 300^\circ) = (6, 300^\circ - 180^\circ) = (6, 120^\circ)$

$(6, -240^\circ)$: same as $(6, 120^\circ)$; $(-6, -60^\circ)$: same as $(-6, 300^\circ)$

$(6, 300^\circ)$ is in QIV:

(c) $x = -6 \cos 300^\circ = -6 \left(\frac{1}{2}\right) = -3$

$y = -6 \sin 300^\circ = -6 \left(-\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$

WHY

Negative r lands in the opposite quadrant.

$(-r, \theta) = (r, \theta \pm 180^\circ).$

$\pm 360^\circ$ keeps the point.

$x = r \cos \theta$; keep r 's sign.

$y = r \sin \theta.$

QII, matches (a).

Give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$. (3 points)

19. $(-2\sqrt{3}, -2)$

WORK

$r = \sqrt{(-2\sqrt{3})^2 + (-2)^2} = \sqrt{12 + 4} = 4$

$\tan \theta = \frac{-2}{-2\sqrt{3}} = \frac{1}{\sqrt{3}}, \theta = 30^\circ$

$\theta = 30^\circ + 180^\circ = 210^\circ$

,

WHY

$r^2 = x^2 + y^2.$

Reference angle.

$x < 0, y < 0$: QIII.

Second pair: $(-r, \theta - 180^\circ).$

Convert each rectangular equation to a polar equation. (2 points each)

20.

WORK

a) $3r \cos \theta - 2r \sin \theta = 6$

$r(3 \cos \theta - 2 \sin \theta) = 6$

b) $r^2 \cos^2 \theta + r^2 \sin^2 \theta = 49$

$r^2(1) = 49, r = \pm\sqrt{49}$

WHY

$x = r \cos \theta, y = r \sin \theta.$

Factor out r .

Solve for r .

Substitute.

$\cos^2 \theta + \sin^2 \theta = 1$; $\pm$ on the square root.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (3 points each)

21. $r = 6 \cos \theta$

WORK

$$r^2 = 6r \cos \theta$$

$$x^2 + y^2 = 6x$$

$$x^2 - 6x + 9 + y^2 = 9$$

$$(x - 3)^2 + y^2 = 9$$

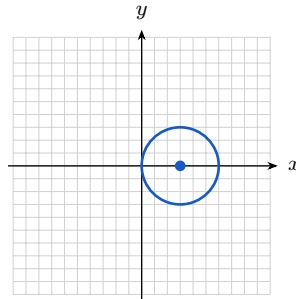
WHY

Multiply both sides by r to make r^2 and $r \cos \theta$.

$$r^2 = x^2 + y^2, r \cos \theta = x.$$

Complete the square: add $(\frac{6}{2})^2$ both sides.

Circle, center $(3, 0)$, radius 3.



Convert each polar equation to a rectangular equation and graph it. (No calculator) (3 points each)

22. $r = \frac{4}{1 + \sin \theta}$

WORK

$$r(1 + \sin \theta) = 4$$

$$r + r \sin \theta = 4, r + y = 4$$

$$r = 4 - y, \sqrt{x^2 + y^2} = 4 - y$$

$$x^2 + y^2 = 16 - 8y + y^2$$

$$x^2 = -8(y - 2)$$

WHY

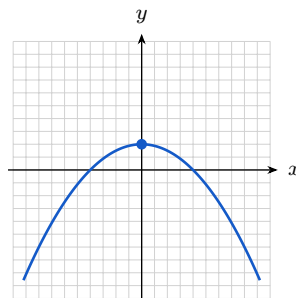
Clear the fraction.

$$r \sin \theta = y.$$

Isolate r , then $r = \sqrt{x^2 + y^2}$.

Square both sides.

Parabola, vertex $(0, 2)$, opens down.



Find the polar equation of the line. Simplify your answer. (3 points)

23. Passes through $(6, 0^\circ)$ and $(4, 90^\circ)$

WORK

$$(6, 0^\circ): x = 6 \cos 0^\circ = 6, y = 6 \sin 0^\circ = 0, (6, 0)$$

$$(4, 90^\circ): x = 0, y = 4, (0, 4)$$

$$m = \frac{4 - 0}{0 - 6} = -\frac{2}{3}, y = -\frac{2}{3}x + 4$$

$$3y = -2x + 12, 2x + 3y = 12$$

$$2r \cos \theta + 3r \sin \theta = 12$$

$$r = \frac{12}{2 \cos \theta + 3 \sin \theta}$$

WHY

Convert both points first.

Slope, then y -intercept 4.

Clear the fraction; standard form.

Substitute.

Factor r , divide.

Find the polar coordinates of the points of intersection for the given interval of θ . (3 points)

24. $r = 2 + 2 \cos \theta$, $r = 3$, $0 \leq \theta < 2\pi$

WORK

$$2 + 2 \cos \theta = 3$$

$$2 \cos \theta = 1, \cos \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$r = 3$ at both

$$\left(3, \frac{\pi}{3}\right), \left(3, \frac{5\pi}{3}\right)$$

WHY

Set $r = r$.

Isolate the trig function.

Cosine positive: QI and QIV.

Plug back in.

$r = 3$ never reaches the pole, so no pole point.

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

WORK

a)

b)

c)

d)

e)

WHY

$n = 3$ odd: 3 petals, each 5 long.

$a + b \cos \theta$ with $\frac{a}{b} = \frac{4}{4} = 1$.

r^2 equation.

$\frac{a}{b} = \frac{2}{5} \neq 1$ (inner loop).

No θ : every point 7 from the pole.